

Nethmi Jayasinghe

Electrical and Computer Engineering, PhD Student

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Profile

PhD student in Electrical and Computer Engineering specializing in reinforcement learning, adaptive decision-making, and robust AI systems. Experienced in Python/PyTorch model development, experimentation pipelines, model evaluation, and SQL-based data analytics. My research focuses on adaptive agents that make real-time decisions under distribution shifts while maintaining stable behavior. Recent work accepted to the International Conference on Machine Learning (ICML), 2026. Interested in applying agentic AI, representation learning, and multi-modal reasoning to real-world search, personalization, recommendation, and interactive AI systems.

Skills

- **Machine Learning & AI:** Reinforcement Learning, Adaptive Decision-Making, Multi-Agent decision systems, Hierarchical RL, Representation Learning, Transformers, Robust Learning, Model Evaluation, Statistical Modeling
- **Deep Learning & Experimentation:** PyTorch, TensorFlow, scikit-learn, Large-scale Training, Experimentation Pipelines, Ablation Studies, Cross-validation, Data Processing, Data Pipelines
- **Programming & Tools:** Python, SQL, Matlab, Linux, Git, MuJoCo, Gymnasium, Webots, Brian2

Industry Experience

Research and Development Engineer at Synopsys Inc - Sri Lanka

06/2022 - 07/2023

- Designed and deployed a scalable SQL-based data pipeline for large-scale test case management, reducing retrieval latency and improving analysis throughput.
- Built an interactive analytics dashboard (Python + SQL) for real-time monitoring of validation metrics, improving team productivity and decision-making.
- Developed automated failure detection and root-cause analysis tools in Python, accelerating debugging workflows and reducing test cycle time.

Publications & Preprints

Residual Control for Fast Recovery from Dynamics Shifts

, Submitted to IROS, 2026

2026

- Designed a real-time adaptive agent system for handling dynamics shifts in interactive environments, enabling rapid recovery without retraining.
- Introduced stability-aligned residual decision mechanisms using magnitude bounds, directional coherence, and performance-conditioned gating to preserve stable behavior during online correction.
- Evaluated across multiple robotic morphologies, reducing recovery time by up to **87%** on Go1, **48%** on Cassie, **30%** on H1, and **20%** on Scout relative to a frozen SAC baseline, with real-world validation on an Agilex Scout Mini Pro.

Inference-Time Residual Adaptation with Cerebellum-Inspired Learning

, International Conference on Machine Learning (ICML), 2026

2025 - 2026

- Developed a PyTorch-based adaptive agent framework that augments a frozen base policy with real-time corrective actions for post-deployment robustness.
- Designed decision-making mechanisms for when and how an agent should intervene, using performance-conditioned gating and dual-timescale adaptation to preserve stable behavior.
- Evaluated agent behavior across MuJoCo locomotion benchmarks and PandaReach-v3 under distribution shifts, achieving up to **+66%** performance improvement on HalfCheetah-v5 and **+53%** on Humanoid-v5 without policy retraining.

Adaptive Reinforcement Learning with Split-Gate Memtransistors

, Submitted to ACS Nano, 2025

2024 - 2025

- Designed a three-agent hierarchical reinforcement learning framework that dynamically modulates update mode (soft vs. hard) and learning rate to improve energy efficiency and convergence.
- Implemented a Deep Q-Learning CartPole agent in Python/PyTorch with supervisory agents for adaptive parameter updates inspired by split-gate memtransistor tunability.
- Demonstrated **6× improvement in rewards** and **70% fewer non-volatile write steps** compared to conventional reinforcement learning baselines.

Fast and Efficient Emergent Differentiation for Novelty Detection Driven by Cerebellum-Inspired Memtransistors

, *Submitted to Nature Communications*, 2025

2024 - 2025

- Implemented a neuromorphic anomaly detection system integrating transformer-based temporal encoding with spiking neural networks and cerebellum-inspired memtransistor synapses.
- Built a Python/PyTorch pipeline for real-time ECG arrhythmia detection, achieving detection within a single heartbeat using **>10⁴× fewer operations** than Transformer baselines.
- Generalized framework to multi-modal inputs (ECG, images, audio) with robustness through ensemble Pkj-like neuron consensus.

Single-Step Extraction of Transformer Attention with Dual-Gated Memtransistor Crossbars

, *IEEE Electron Device Letters*, 2024

2023 - 2024

- Developed a hardware–software co-design framework to accelerate Transformer attention mechanisms using dual-gated memtransistor crossbars.
- Eliminated query/key computation by enabling direct attention score extraction, reducing operations and memory overhead.
- Achieved **2.37× lower energy consumption** with fewer crossbar cells compared to conventional compute-in-memory approaches, validated on multiple time-series datasets.

Education

University of Illinois at Chicago

USA 2023-Present

Ph.D. in Electrical and Computer Engineering

Expected Graduation: 05/2028

GPA: 3.9/4.0

Relevant Coursework: Reinforcement Learning, Neural Networks, Computer Vision, Pattern Recognition, Information & Learning, Control Engineering, Algorithms

University of Moratuwa

Sri Lanka 2017-2022

B.Sc. (Hons) in Electrical Engineering

GPA: 3.81/4.2